

Sangwon Lee

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ACADEMIC INTERESTS

I am interested in designing and analyzing algorithms for real-world problems from a theoretical perspective, developing methods that are both practically effective and theoretically justified. My current interests lie in machine learning theory and sequential decision making, particularly in understanding how theoretical insights can inform the design of effective learning algorithms for real-world applications.

EDUCATION

Seoul National University Mar. 2020 – Present
B.S. Aerospace Engineering; Double Major in Computer Science & Engineering and Mathematics Seoul, Korea
GPA: 3.98 / 4.3 (overall) 4.05 / 4.3 (major)

Republic of Korea Army Feb. 2022 – Aug. 2023
Mandatory Military Service, Honorably Discharged as Sergeant

Gyeonggi Science High School Mar. 2017 – Feb. 2020

RESEARCH & PROJECTS

Undergraduate Thesis Research Mar. 2025 – Dec. 2025
Supervised by Prof. Hyunjin Kim, Seoul National University

- Developed an Edge Semantic Injection Layer (Edge-SIL) integrating LLM-derived predicate representations via cross-attention, enabling structured relational inference under open-vocabulary settings.
- Designed context-aware prompting strategies for long-tail relation prediction and evaluated performance on the 3DSSG benchmark.

Distributed Optimization for LLM Inference Fall 2025
in lecture: Scalable High-Performance Computing

- Optimized inference of an 8B-parameter LLM across multi-node CPU/GPU systems using MPI, OpenMP, SIMD, and CUDA; analyzed memory/compute bottlenecks to improve throughput.

ARC-AGI Reasoning Challenge Spring 2025
in lecture: Introduction to Deep Learning

- Designed learning strategies for few-shot abstract reasoning; achieved 1st place in the course-wide competition.

HONORS, AWARDS & SCHOLARSHIPS

1st Place (AI Sector), Hanwha Life Future Talent Contest 2026

Full Tuition Waiver Scholarship, Seoul National University Spring 2024

Boeing Korea Scholarship Fall 2021

Excellent Award (1st Place), R&E Joint Presentation Competition (*awarded by President of GIST*) 2019

International Physics Olympiad (IPhO) Winter School: Completed 2018

SKILLS

Mathematical Background: Measure Theory, Machine Learning Theory, Statistical Machine Learning, Optimization

Languages: Python, C/C++, MATLAB, $\text{\LaTeX}$

ML / Systems: PyTorch, CUDA, MPI